

Notes: Demyanyk & Van Hemert (RFS, 2011)

Understanding the Subprime Mortgage Crisis

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1 Big picture

- The paper uses LoanPerformance loan-level data to diagnose causes of the subprime mortgage crisis.
- It finds that adjusted loan quality deteriorated monotonically from 2001 to 2007.
- Observed loan, borrower, and macro variables cannot fully explain poor 2006–2007 performance.
- High house-price appreciation in 2003–2005 masked declining underwriting quality.
- Securitizers partly recognized high-LTV risk, but aggregate subprime-prime spreads declined.
- The paper interprets the episode as a classic lending boom-bust: rapid growth, deteriorating standards, rising risk, and falling risk premia.

2 Incremental contribution of the paper

- **Loan-level decomposition.** Quantifies how much poor performance can be explained by observables.
- **Adjusted vintage-quality measure.** Constructs adjusted delinquency curves controlling for loan, borrower, and macro characteristics.
- **Early-warning result.** Shows that deterioration was visible by the end of 2005.
- **Securitizer-awareness result.** Shows mortgage rates became more sensitive to LTV risk over time.
- **Risk-premium compression.** Shows adjusted subprime-prime spreads declined despite rising adjusted risk.
- **LMI-neighborhood evidence.** Shows higher delinquency risk in low- and moderate-income neighborhoods after controls.

3 Data and measurement

The data come from First American CoreLogic LoanPerformance and cover about 85% of securitized subprime mortgages, more than half of the U.S. subprime market. The sample covers 2001–2007 first-lien subprime loans. The main outcome is first-time delinquency: a loan sixty or more days past due, in foreclosure, REO, or default. The appendix repeats the analysis using foreclosure.

4 Models and empirical specifications

4.1 Actual delinquency rate

$$Actual_{k,t} = 1 - \prod_{s=1}^t (1 - \tilde{P}_{s,k}).$$

4.2 Proportional odds model

$$logit(P_{i,s}) = \alpha_s + \beta' X_i + \gamma_{vintage(i)}.$$

Vintage coefficients measure adjusted loan quality after controlling for FICO, LTV, debt-to-income, documentation, product type, mortgage rate, local house prices, unemployment, and neighborhood income.

4.3 Mortgage-rate sensitivity to LTV

$$Rate_{i,t} = \alpha_t + b_{LTV,t}LTV_i + Controls_i + \varepsilon_{i,t}.$$

Rising $b_{LTV,t}$ implies that securitizers became more sensitive to high-LTV risk.

4.4 Subprime-prime spread regression

$$Spread_{i,t} = \beta_0 + \beta_1 PrimeRate_t + \beta_2' Characteristics_i + error_{i,t}.$$

The prediction error measures whether subprime rates are high or low relative to risk characteristics.

5 Identification and empirical design

The paper is a diagnostic decomposition rather than a policy experiment. Identification comes from comparing vintages at the same loan ages while controlling for observed borrower, loan, and macro characteristics. The key question is whether later vintages perform worse even after adjustment. They do. The early-warning exercise uses data only through 2005 and still finds deterioration. Pricing regressions test whether lenders recognized growing LTV risk. Limitations include focus on securitized subprime loans, MSA-level house-price measures, and unobserved underwriting variables.

6 Section-by-section study guide

- **Introduction.** Contribution roadmap and boom-bust interpretation.
- **Descriptive analysis.** Loan growth, product mix, documentation decline, and raw delinquency patterns.
- **Empirical strategy.** Proportional odds duration model and adjusted vintage quality.
- **Baseline results.** Determinants of delinquency and monotonic deterioration in adjusted quality.
- **Robustness.** Alternative controls, risk layering, and early-warning analysis.
- **LTV nonstationarity.** High-LTV borrowers become increasingly risky.
- **Spread analysis.** Risk premia decline despite rising risk.

7 Tables and figures

7.1 Figure 1: Actual and adjusted delinquency rate + Table 1: Loan characteristics at origination for different vintages

Read: Actual delinquency is highest for 2007, 2006, and 2001. Adjusted delinquency rises monotonically from 2001 to 2007. Table 1 shows rapid loan growth, declining documentation, rising CLTV, and changes in mortgage type.

Detailed interpretation: The adjusted panel shows that 2001 looked bad because of unfavorable observable conditions, while 2006–2007 were intrinsically poor after adjustment.

Economic message: The crisis was rooted in deteriorating loan quality years before 2007.

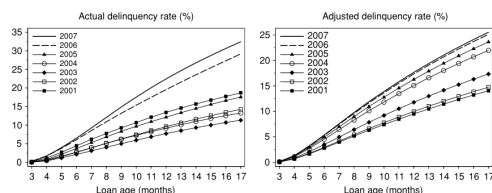


Figure 1
Actual and adjusted delinquency rate
The figure shows the age pattern in the actual (left panel) and adjusted (right panel) delinquency rate for the different vintage years. The delinquency rate is defined as the cumulative fraction of loans that were past due sixty or more days, in foreclosure, real-estate owned, or defaulted, at or before a given age. The adjusted delinquency rate is obtained by adjusting the actual rate for each by year variation in FICO score. LTV

(a) Figure 1: actual and adjusted delinquency rate

Table 1
Loan characteristics at origination for different vintages

	2001	2002	2003	2004	2005	2006	2007
<i>Size</i>							
Number of Loans (*1000)	452	737	1,258	1,911	2,274	1,772	316
Average Loan Size (*1000)	126	145	164	180	200	212	220
<i>Mortgage type</i>							
FRM (%)	33.2	29.0	33.6	23.8	18.6	19.9	27.5
ARM (%)	0.4	0.4	0.3	0.3	0.4	0.4	0.2
Hybrid (%)	59.9	68.2	65.3	75.8	76.8	54.5	43.8
Balloon (%)	6.5	2.5	0.8	0.2	4.2	25.2	28.5
<i>Loan purpose</i>							
Purchase (%)	29.7	29.3	30.1	35.8	41.3	42.4	29.6
Refinancing (cash out) (%)	58.4	57.4	57.7	56.5	52.4	51.4	59.0
Refinancing (no cash out) (%)	11.2	12.9	11.8	7.7	6.3	6.2	11.4
<i>Variable means</i>							
FICO Score	601.2	608.9	618.1	618.3	620.9	618.1	613.2
Combined LTV Ratio (%)	79.4	80.1	82.0	83.6	84.9	85.9	82.8
Debt-to-Income Ratio (%)	38.0	38.5	38.9	39.4	40.2	41.1	41.4
Missing Debt-to-Income Ratio Dummy (%)	34.7	37.5	29.3	26.5	31.2	19.7	30.9
Investor Dummy (%)	8.2	8.1	8.1	8.3	8.3	8.2	8.2
Documentation Dummy (%)	76.5	70.4	67.8	66.4	63.4	62.3	66.7
Prepayment Penalty Dummy (%)	75.9	75.3	74.0	73.1	72.5	71.0	70.2
Mortgage Rate (%)	9.7	8.7	7.7	7.3	7.5	8.4	8.6
Margin for ARM and Hybrid Mortgage Loans (%)	6.4	6.6	6.3	6.1	5.9	6.1	6.0

Descriptive statistics for the first-lien subprime loans in the Loan Performance database.

1.1 Loan characteristics at origination

Table 1 provides the descriptive statistics for the subprime mortgage loans in our database that were originated between 2001 and 2007. In the first block of Table 1, we see that the annual number of originated loans increased by a factor of four between 2001 and 2006 and the average loan size almost doubled over those five years. The total dollar amount originated in 2001 was \$87

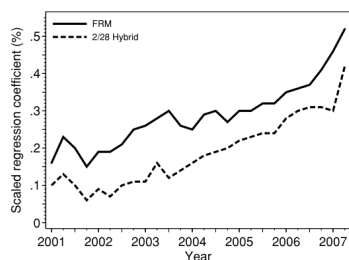
(b) Table 1: loan characteristics at origination

7.2 Figure 2: Sensitivity of mortgage rate to first-lien LTV + Table 2: Baseline-case variable definitions

Read: Mortgage rates became more sensitive to LTV over time; Table 2 defines the risk covariates.

Detailed interpretation: Securitizers saw high-LTV borrowers becoming riskier, but overall risk premia later fell.

Economic message: Awareness of some risks did not prevent underpricing of aggregate risk.



(a) Figure 2: LTV sensitivity

Table 2
Baseline-case variable definitions

Variable (expected sign)	Explanation
FICO Score (+)	Fair Isaac and Company (FICO) credit score at origination.
Combined LTV (+)	Combined value of all liens divided by the value of the house at origination. A higher combined LTV ratio makes debt attractive.
Debt-to-Income Ratio (+)	Back-end debt-to-income ratio, defined by the total monthly debt payments divided by the gross monthly income, at origination. A higher debt-to-income ratio makes it harder to make the monthly mortgage payment.
Missing Debt-to-Income Dummy (+)	Equals one if the back-end debt-to-income ratio is missing and zero if provided. We expect the lack of debt-to-income info to be a negative signal on borrower quality.
Cash-Out Dummy (+)	Equals one if the mortgage loan is a cash-out refinancing loan. Pennington-Cross and Chomsisengphet (2007) show that common reasons to initiate a cash-out refinancing are to consolidate debt and to improve property.
Investor Dummy (+)	Equals one if the borrower is an investor and does not owner-occupy the property.
Documentation Dummy (+)	Equals one if full documentation on the loan is provided and zero otherwise. We expect full documentation to be a positive signal on borrower quality.
Prepayment Penalty Dummy (+)	Equals one if there is a prepayment penalty and zero otherwise. We expect that a prepayment penalty makes refinancing less attractive.
Mortgage Rate (+)	Initial interest rate as of the first payment date. A higher interest rate makes it harder to make the monthly mortgage payment.
Margin (+)	Margin for an adjustable-rate or hybrid mortgage over an index interest rate, applicable after the first interest rate reset. We consider four product types: FRMs, Hybrids, ARMs, and Balloons. We include a dummy variable for the latter three types because they have the interpretation of the probability of delinquency relative to FRM. Because we expect the FRM to be of more risk-averse and prudent borrowers, we expect positive signs for all three product type dummies.
Product Type Dummies (+)	Size of the mortgage loan. We have no clear prior on the effect of the origination amount on the probability of delinquency constant the LTV and debt-to-income ratios.
Origination Amount (?)	MSA-level house price appreciation from the time of loan origination, reported by the Office of Federal Housing Enterprise (OFHEO). Higher housing equity leads to better opportunities to refinance the mortgage loan.
House Price Appreciation (+)	State-level change in the unemployment rate from the time of loan origination, reported by the Bureau of Economic Analysis. An increase in the state unemployment rate increases the probability a homeowner lost his job, which increases the probability of delinquency.
Change Unemployment Rate (+)	ZIP Code level median income in 1999 from the U.S. Census Bureau 2000. The better the neighborhood, as proxied by its income, the more motivated a borrower may be to stay current on a mortgage.
Neighborhood Income (+)	

This table presents definitions of the baseline-case variables (covariates) used in the proportional odds duration model. The first two variables are used as dependent variables. TI variables are used as independent variables. We report the expected sign for the independent variables in parentheses and sometimes provide a brief motivation.

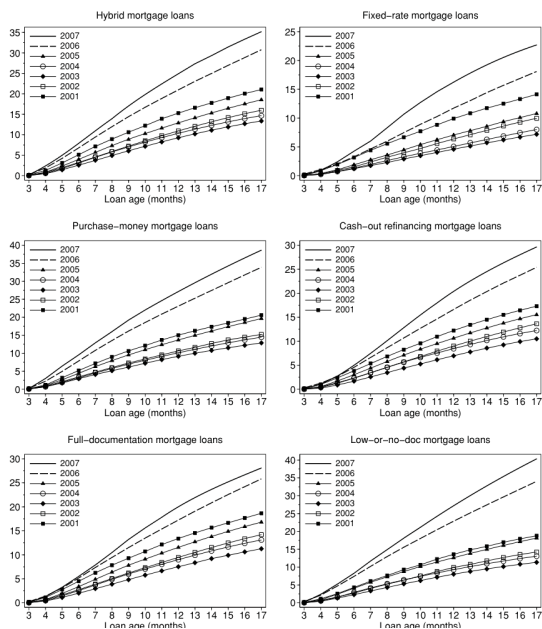
(b) Table 2: variable definitions

7.3 Figure 3: Actual delinquency by segment + Figure 4: Outstanding mortgage delinquency

Read: Later vintages perform worse across contract type, loan purpose, and documentation categories; outstanding-loan delinquency rates can mask vintage deterioration.

Detailed interpretation: The deterioration is not just due to one product such as hybrid mortgages.

Economic message: Vintage-level loan-age curves are necessary to see worsening underwriting.



(a) Figure 3: market segments

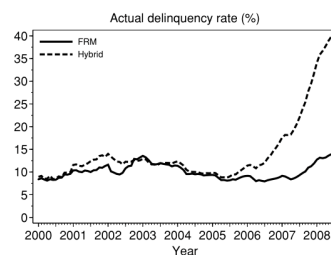


Figure 4
Actual delinquency rates of outstanding mortgages
The figure shows the actual delinquency rates of all outstanding FRMs and hybrids from January 2000 through

(b) Figure 4: outstanding mortgages

7.4 Figure 5 and Tables 3–5

Read: Figure 5 shows deterioration visible by 2005. Tables 3–5 report covariate, vintage, and age effects in the proportional odds model.

Detailed interpretation: Observables explain cross-sectional risk, but vintage effects still deteriorate monotonically.

Economic message: Problems were detectable before the crisis.

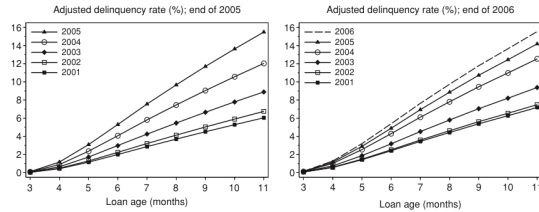


Figure 5
Adjusted delinquency rate, viewed at the end of 2005 and 2006
The figure shows the adjusted delinquency rate using data available at the end of 2005 (left panel) and 2006 (right panel). The delinquency rate is defined as the cumulative fraction of loans that were past due sixty or more days, in foreclosure, real-estate owned, or defaulted, at or before a given age. The adjusted delinquency rate is obtained by adjusting the actual rate for year-by-year variation in FICO scores, LTV ratios, debt-to-income ratios, missing

(a) Figure 5: early-warning adjusted delinquency

Table 4
Determinants of delinquency, vintage dummy variables

Explanatory variable	Estimate
Vintage 2001	0.00
Vintage 2002	5.12 ^a
Vintage 2003	23.64 ^a
Vintage 2004	49.71 ^a
Vintage 2005	57.93 ^a
Vintage 2006	64.65 ^a
Vintage 2007	66.69 ^a

The table shows the output of the proportional odds duration model. The first column reports the vintage dummies included. The values for the other covariates are outlined in Tables 3 and 5. The second column shows the estimated coefficients. The vintage 2001 dummy was not included as covariate.

(a) Table 4: vintage effects

Table 3
Determinants of delinquency, variables other than vintage and age dummy variables

Explanatory variable	Marginal effect, %	Contribution, %				
	2001–2007	2001	2002	2003	2004	20
FICO Score	−47.94 ^a	13.74	6.74	−0.36	−1.00	−3.
Combined LTV Ratio	24.02 ^a	−7.44	−5.92	−2.52	0.23	2.
Debt-to-Income Ratio	12.86 ^a	−2.76	−2.73	−0.66	0.31	−0.
Missing Debt-to-Income Dummy	12.59 ^a	2.17	2.35	0.24	−0.51	0.
Cash-Out Dummy	−12.73 ^a	−0.98	−0.60	−0.69	−0.35	0.
Investor Dummy	3.86 ^a	0.04	−0.03	−0.01	0.02	0.
Documentation Dummy	−13.79 ^a	−3.42	−1.49	−0.54	0.01	0.
Prepayment Penalty Dummy	6.13 ^a	0.50	0.37	0.18	0.05	−0.
Mortgage Rate	29.21 ^a	39.51	16.86	−4.78	−13.00	−8.
Margin	11.84 ^a	−2.07	0.19	−1.81	0.32	1.
Hybrid Dummy	1.81	−0.23	0.03	−0.12	0.29	0.
ARM Dummy	0.36	0.00	0.00	−0.01	0.00	0.
Balloon Dummy	3.70 ^a	−0.16	−0.71	−0.94	−1.02	−0.
Origination Amount	15.91 ^a	−7.31	−5.03	−2.64	−0.62	2.
House Price Appreciation	−29.96 ^a	7.24	5.32	−10.52	−16.40	−2.
Change Unemployment	7.69 ^a	12.76	3.03	−1.85	−2.40	−2.
Neighborhood Income	−8.81 ^a	0.11	−0.40	−0.53	−0.11	0.
Total	—	51.71	17.99	−27.55	−34.19	−9.

(b) Table 3: covariate effects

Table 5
Determinants of delinquency, age dummy variables

	Age (months)							
	3	4	5	6	7	8	9	10
AgeEffect (*100)	0.04	0.31	0.55	0.67	0.72	0.72	0.73	0.7
	Age (months)							
	11	12	13	14	15	16	17	—
AgeEffect (*100)	0.73	0.72	0.73	0.71	0.70	0.69	0.67	—

The table shows the output of the proportional odds duration model. Each column reports the corresponding odds ratio, *AgeEffect*, estimated for each age dummy variable based on Equation (6). The values for the other covariates are reported in Tables 3 and 4.

(b) Table 5: age effects

7.5 Table 6: Low- and moderate-income neighborhood effects

Read: Low neighborhood income and LMI indicators raise delinquency risk after controls.

Detailed interpretation: The table suggests a possible negative by-product of credit expansion into targeted neighborhoods.

Economic message: Credit access policy can interact with weaker underwriting.

Table 6
Low- and moderate-income neighborhood effects

	I	II	III	IV
Neighborhood Income	−8.81%*	−9.10%*	−6.95%*	−6.94%*
MSA Income	—	1.07%*	—	—
LMI Dummy	—	—	2.82%*	—
LMI 2001	—	—	—	6.26%*
LMI 2002	—	—	—	4.52%*
LMI 2003	—	—	—	1.75%
LMI 2004	—	—	—	3.64%*
LMI 2005	—	—	—	2.67%*
LMI 2006	—	—	—	1.79%*
LMI 2007	—	—	—	3.79%*

This table reports marginal effects for different measures of neighborhood (median household) income relative to MSA (median household) income. In all four specifications, we include the baseline case variables used in Table 3, of which we only report neighborhood income to preserve space. Specification I is the baseline case with no variable capturing the neighborhood income relative to the MSA income. Specification II includes MSA income. Specification III includes a dummy with value one if neighborhood income is less than 80% of MSA income; i.e., if it is a low- and moderate-income (LMI) neighborhood. Specification IV includes interaction variables of the LMI dummy with vintage year dummies. The marginal effect for the interaction terms is computed as the product of the estimated coefficient and the full sample standard deviations of LMI. * indicates statistical significance at the 1% level.

Table 6: LMI neighborhood effects

7.6 Figure 6 and Figure 7: Spread evidence

Read: Subprime-prime spreads and adjusted prediction errors decline over time.

Detailed interpretation: Risk premia compressed while adjusted loan risk increased.

Economic message: The boom featured declining compensation for rising risk.

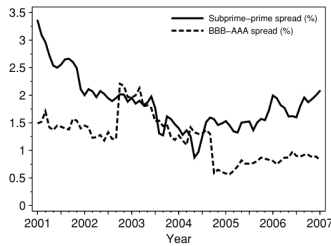
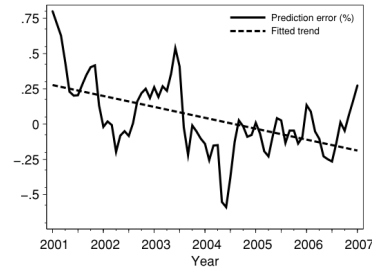


Figure 6

(a) Figure 6: rate spread



(b) Figure 7: spread prediction error

7.7 Appendix figures: Figures 8–10

Read: Foreclosure results, separate FRM/hybrid estimates, and nonlinear LTV pricing all support the baseline conclusions.

Detailed interpretation: The main deterioration result is robust to outcome definition, contract type, and nonlinear risk pricing.

Economic message: The boom-bust diagnosis is robust.

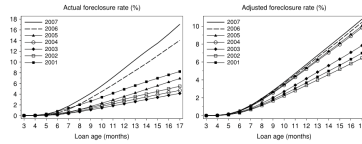


Figure 8
Actual and adjusted foreclosure rates
The figure shows the age pattern in the actual (left panel) and adjusted (right panel) foreclosure rate for the different vintage years. The foreclosure rate is defined as the cumulative fraction of loans that were in foreclosure, real-

(a) Figure 8: foreclosure

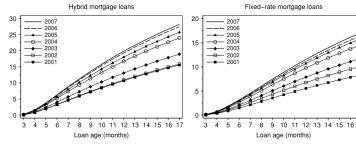


Figure 9
Adjusted delinquency rates for hybrids and FRMs separately
The figure shows the adjusted delinquency rates based on hybrid mortgages (left panel) and FRMs (right panel) separately. The delinquency rate is defined as the cumulative fraction of loans that were past due sixty or more

(b) Figure 9: hybrids and FRMs

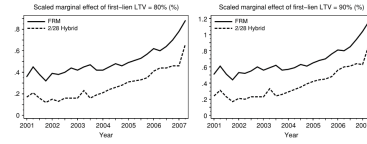


Figure 10
Sensitivity of mortgage rate to first-lien LTV ratio allowing for nonlinearity
The figure shows the scaled marginal effect of the first-lien LTV ratio on the mortgage rate for first-lien fixed-rate and 238 hybrid mortgages, evaluated at a first-lien LTV of 80% (left panel) and 90% (right panel). The effect

(c) Figure 10: nonlinear LTV sensitivity

8 Short summary

- Loan-level subprime data show adjusted loan quality deteriorating from 2001 to 2007.
- Observed borrower and loan characteristics cannot fully explain the poor performance of 2006–2007 loans.
- High house-price appreciation masked deteriorating underwriting during 2003–2005.
- High-LTV borrowers became increasingly risky, and mortgage rates became more sensitive to LTV.
- But subprime-prime spreads declined, meaning compensation for risk fell.
- The evidence fits a lending boom-bust scenario.

9 Conclusion

9.1 Core conclusion

Subprime loan quality deteriorated long before the crisis, and this deterioration was visible before 2007.

9.2 Economic intuition

Rising house prices reduced realized defaults and allowed weak underwriting to remain hidden. When appreciation stopped, risk was revealed.

9.3 Incremental contribution

The paper decomposes crisis performance at the loan level, constructs adjusted vintage-quality curves, detects early-warning signals, studies securitizer awareness, and shows risk-premium compression.

9.4 Policy implication

Supervisors and investors should monitor adjusted loan-quality measures and risk premia, not just recent raw default rates.

9.5 Limitations

The data cover securitized subprime loans and cannot observe all underwriting incentives, fraud, or unmeasured risk.

9.6 Takeaway

The crisis was the culmination of a boom in which lending standards deteriorated and compensation for risk fell.